

Math 142, S12

Quiz 4

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (6 points) Determine whether the following improper integrals are convergent or divergent and evaluate those that are convergent.

(a) $\int_2^{+\infty} e^{-x/2} dx$

$$= \lim_{t \rightarrow +\infty} \left(\int_2^t e^{-x/2} dx \right)$$

$$= \lim_{t \rightarrow +\infty} (-2e^{-x/2}) \Big|_2^t = \lim_{t \rightarrow +\infty} ([-2e^{-t/2}] - [-2e^{-1}])$$

$$= \lim_{t \rightarrow +\infty} (-2e^{-t/2} + \frac{2}{e})$$

$$= 0 + \frac{2}{e} = \frac{2}{e} \quad \text{Thus it is convergent.}$$

(b) $\int_{-\infty}^{2\pi} \cos \theta d\theta$

$$= \lim_{t \rightarrow -\infty} \left(\int_t^{2\pi} \cos \theta d\theta \right)$$

$$= \lim_{t \rightarrow -\infty} (\sin \theta \Big|_t^{2\pi})$$

$$= \lim_{t \rightarrow -\infty} (0 - \sin t)$$

$$= - \lim_{t \rightarrow -\infty} \sin t \rightarrow DNE$$

Thus it is divergent.

Problem 2. (4 points) Evaluate the integral

$$\int e^{\sqrt{x}} dx.$$

(Hint: First use the substitution $u = \sqrt{x}$.)

Solution: $u = \sqrt{x} \rightarrow u^2 = x \rightarrow 2u du = dx$ ← substitution.

then $\int e^{\sqrt{x}} dx \xrightarrow{u=\sqrt{x}} \int e^u 2u du = 2 \int u e^u du$

$$\int u e^u du \xrightarrow{\uparrow} u e^u - \int e^u du = u e^u - e^u$$

← integration
by parts

$$\begin{aligned} f(u) &= u, & g'(u) &= e^u \\ f'(u) &= 1, & g(u) &= e^u \end{aligned}$$

$$= e^u(u-1)$$

Thus

$$\int e^{\sqrt{x}} dx = 2e^{\sqrt{x}}(\sqrt{x}-1) + C$$